

COMPUTO EN LA NUBE

INGENIERIA DE CIBERSEGURIDAD E
INFRAESTRUCTURA DE COMPUTO



Virtualización

- Hypervisores
- Máquinas virtuales
- Virtualización de dispositivos
- Administración de servicios virtualizados
- Virtualización de red
- Virtualización de almacenamiento



Introducción

- ¿Qué es la virtualización?
- Virtual según la RAE:
Que está ubicado o tiene lugar en línea, generalmente a través de internet. *Tienda, campus, curso, encuentro virtual*.
- Es la creación a través de software de una versión virtual de algún recurso tecnológico (Wikipedia)
- Es la abstracción de los recursos de una computadoras /
La virtualización es utilizada para abstraer el sistema operativo y las aplicaciones de un hardware para construir un ambiente de servidores más eficiente, ágil y simple (IEEE)
- Virtualización es la simulación del software y/o hardware sobre el que se ejecuta otro software (NIST)



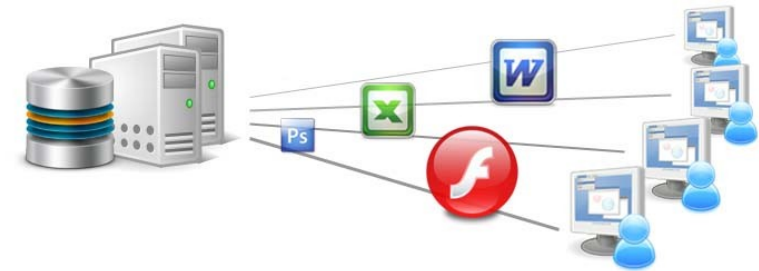
Tipos de virtualizació



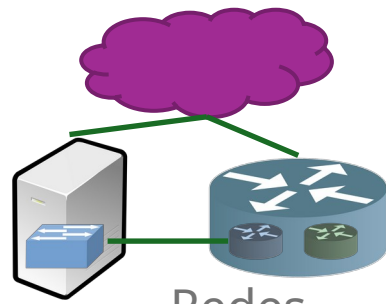
Servidor
es



Escritori
os



Aplicacion
es



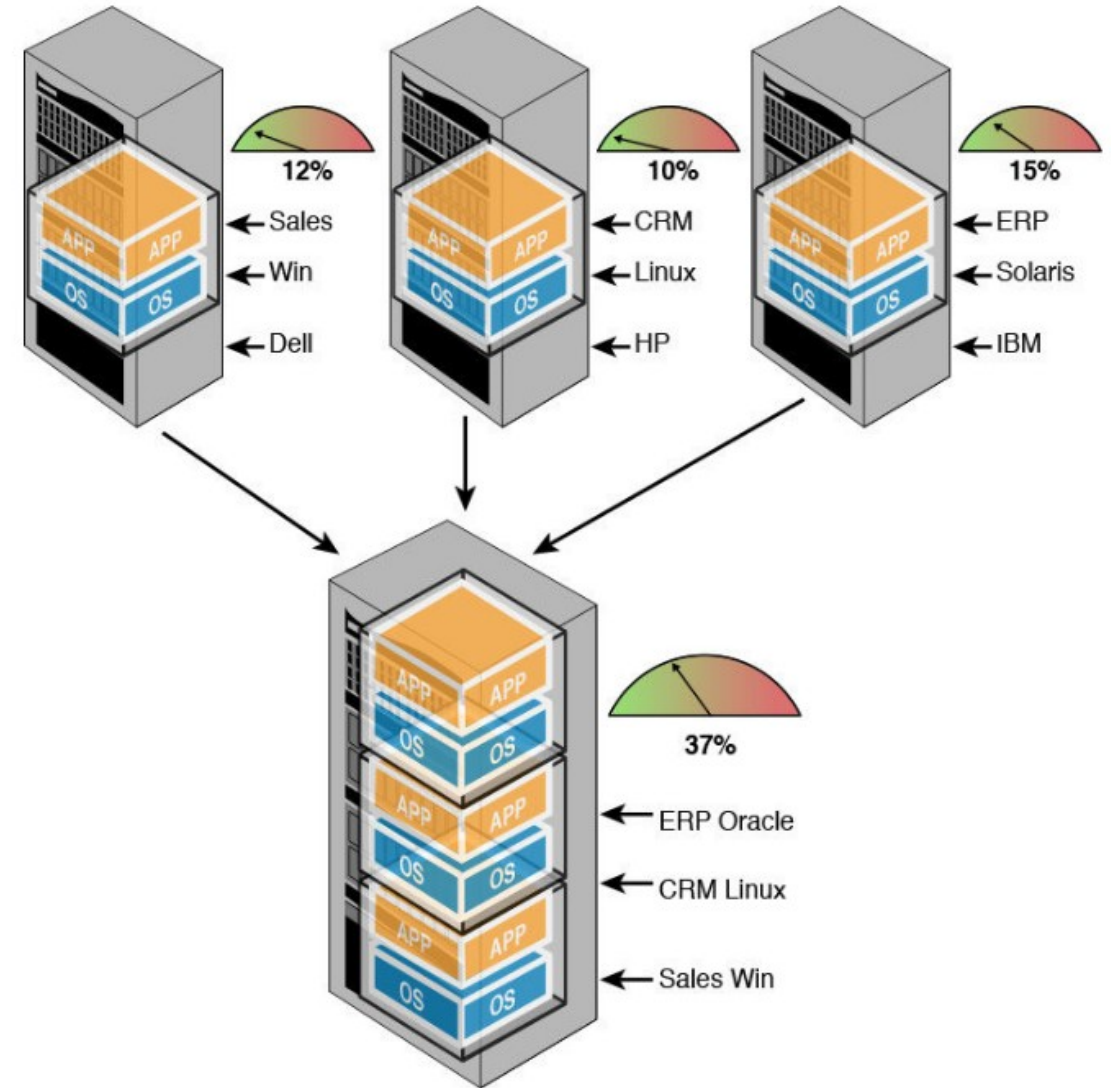
Redes



Almacenamien
to

¿Por qué virtualizar?

- Crecimiento del uso de TI en todas las actividades
- SLA's cada vez más elevados
- Subutilización de infraestructura
- Costos de sistemas auxiliares en centros de datos
- Costos de cableado
- Costos de administración
- Velocidad de innovación
- Recuperación de desastres



Hipervisor

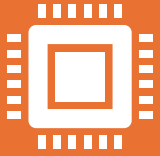
Es una plataforma que permite aplicar diversas técnicas de control de virtualización para utilizar, al mismo tiempo, diferentes sistemas operativos en una misma computadora (Wikipedia)

Es un software que crea y ejecuta máquinas virtuales (VM) y que, además, aísla su sistema operativo y recursos de las máquinas virtuales y permite crearlas y gestionarlas. (Red Hat)

Funciones

- **Sistema operativos de sistemas operativos**
 - Se refiere a un software que permite a los dispositivos de hardware de un equipo de cómputo compartirse entre distintas máquinas virtuales.
- **Monitor de máquinas virtuales**
 - Coordinar las peticiones de acceso y flujos de información entre las máquinas virtuales y los recursos de hardware y viceversa

Técnicas de virtualización



Virtualización completa
utilizando traducción binaria.

Se basa en traducir el código del kernel para reemplazar las instrucciones no virtualizables con nuevas secuencias de instrucciones, que tienen el efecto deseado en el hardware.

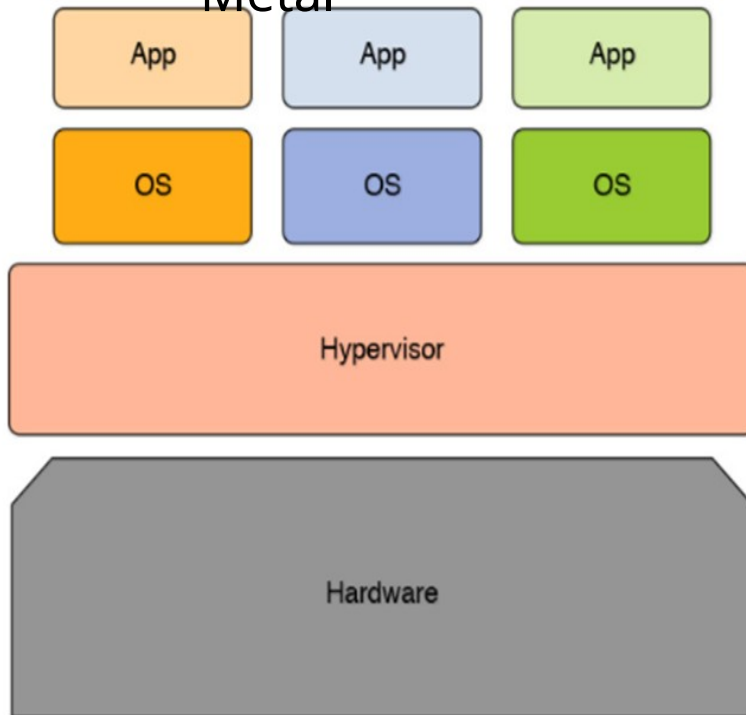


Virtualización asistida por
sistema operativo o
paravirtualización.

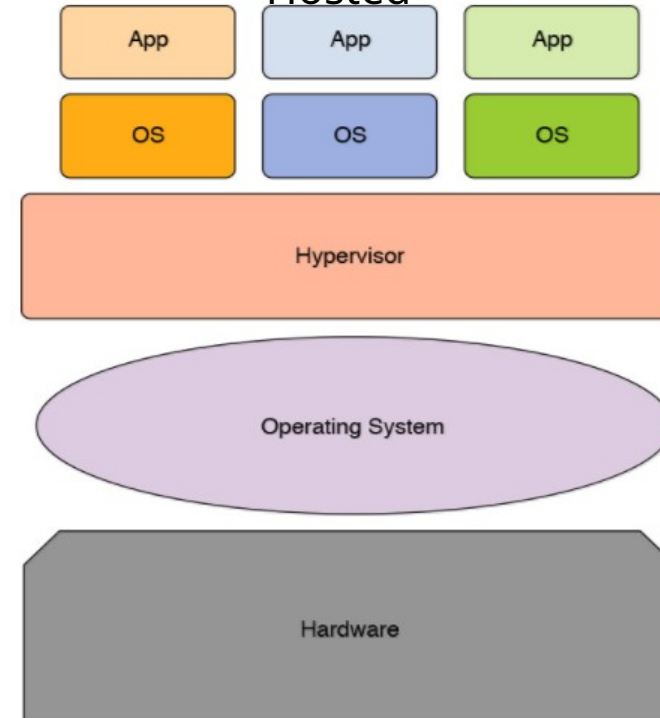
Implica modificar el kernel del SO huésped para reemplazar las instrucciones no virtualizables con -hypercalls-, las que se comunican directamente con el hypervisor de la capa de virtualización

Tipos de hipervisores

Tipo 1 - Bare Metal



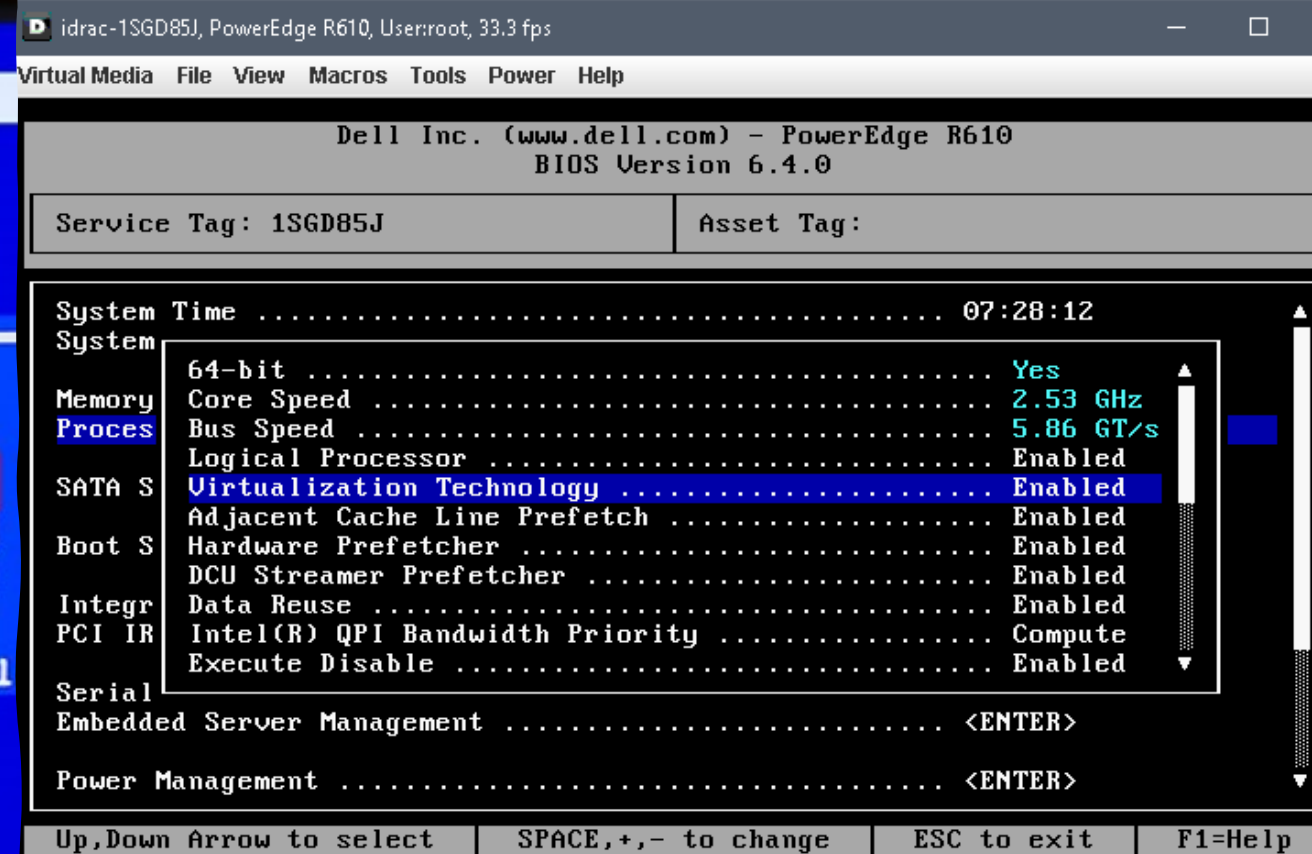
Tipo 2 - Hosted



A glowing blue microprocessor is shown in a close-up, angled view. The processor is a square chip with a dark center and a lighter, textured border. It is mounted on a blue circuit board with visible traces and components. The entire scene is illuminated with a strong blue light, creating a futuristic and high-tech atmosphere.

Virtualización asistida por hardware

- Multitasking, Multicore
- Uno de los detonadores más importantes para la virtualización, fueron las tecnologías de virtualización asistida por hardware en los procesadores (**2005**)
 - Intel VT
 - AMD-V
- Agrega extensiones
- Puede combinarse con las anteriores



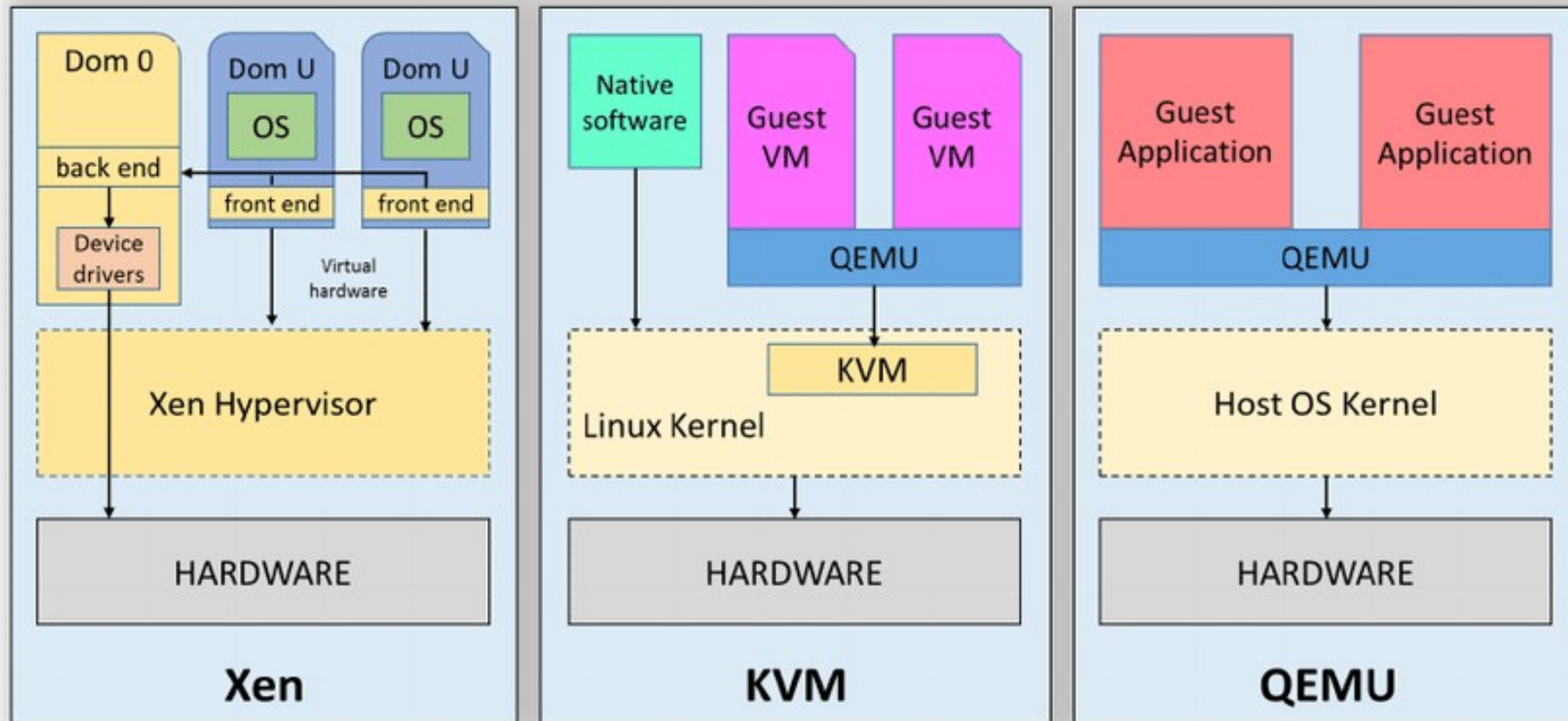
BIOS

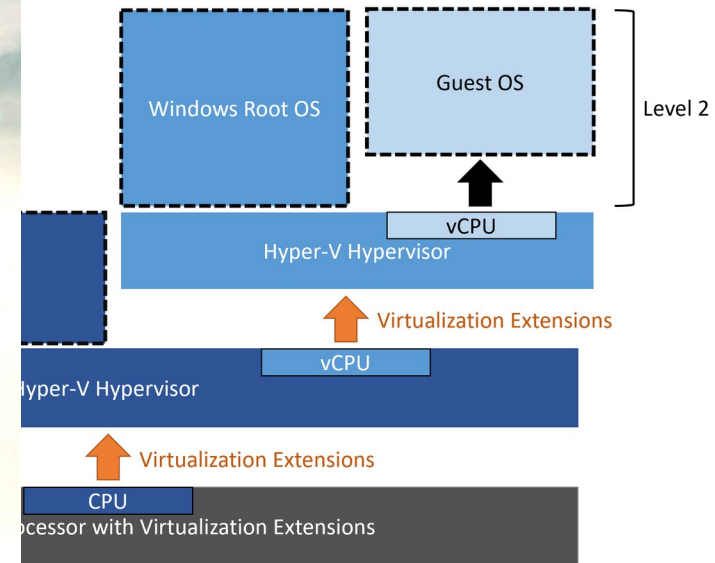
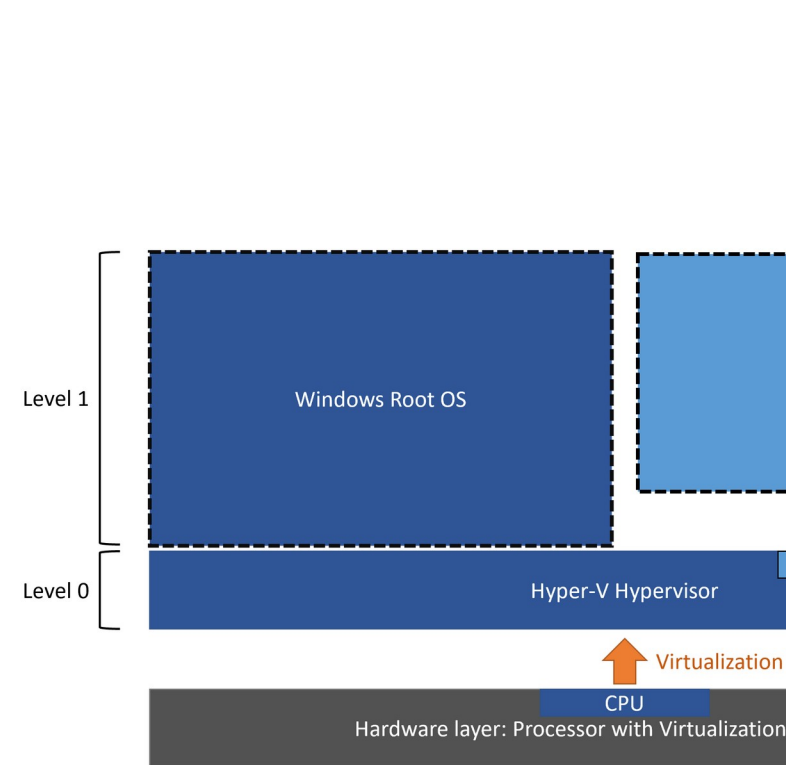
Activación de funciones de virtualización

Hipervisor	Descripción	Tipo
Xen	Hipervisor open source	Tipo 1
Vmware ESXi vSphere	Hipervisor para datacenter	Tipo 1
Vmware Fusion – Horizon FLEX - Workstation	Hipervisor para desktop	Tipo 2
Hyper-V	Hipervisor de Microsoft	Tipo 1
VirtualBox	Hipervisor de Oracle	Tipo 2
Oracle VM	Hipervisor de Oracle para datacenter	Tipo 1

Característica	VirtualBox	Xen	KVM	VMware Workstation	VMware vSphere (ESXi)	Hyper-V	Proxmox VE
Tipo de Hipervisor	Tipo 2 (Hosted)	Tipo 1 (Bare-metal)	Tipo 1 (Kernel integrado)	Tipo 2 (Hosted)	Tipo 1 (Bare-metal)	Tipo 1 (Bare-metal)	Tipo 1 (Bare-metal)
Plataforma Base	Windows, Linux, macOS	Linux (con Dom0)	Linux (Kernel Linux)	Windows, Linux	VMware ESXi	Windows Server	Linux (Debian-based)
Arquitectura	Basado en software	Hipervisor liviano	Kernel integrado	Basado en software	Hipervisor propietario	Hipervisor nativo de Windows	Basado en KVM y LXC
Requiere SO Host?	Sí	No (usa Dom0)	No	Sí	No	No	No
Soporte de SO Invitado	Windows, Linux, macOS	Windows, Linux, BSD	Windows, Linux, BSD	Windows, Linux, BSD	Windows, Linux, BSD	Windows, Linux, BSD	Windows, Linux, BSD
Gestión de VM	GUI y CLI (VBoxManage)	CLI, XenCenter	CLI, Virt-Manager	GUI	vCenter, ESXi CLI	Hyper-V Manager, PowerShell	GUI y CLI (Web UI, qm CLI)
Rendimiento	Bajo-Medio	Alto	Alto	Medio	Muy alto	Alto	Alto
Snapshot/Clonación	Sí	Sí	Sí	Sí	Sí	Sí	Sí
Migración en Vivo	No	Sí	Sí	No	Sí	Sí	Sí
Soporte de Contenedores	No	No	Sí (con LXC)	No	No	Sí (con WSL y Docker)	Sí (LXC y KVM)
Casos de Uso	Pruebas, desarrollo	Cloud, servidores	Servidores, cloud	Pruebas, desarrollo	Data centers, cloud	Infraestructura empresarial	Virtualización empresarial y homelab
Costo	Gratis	Gratis (Open Source)	Gratis (Open Source)	Comercial (de pago)	Comercial (de pago)	Incluido en Windows Server	Gratis (con versión de pago para soporte)

Comparativa





Virtualiz

yper-V

ORACLE[®]

VM



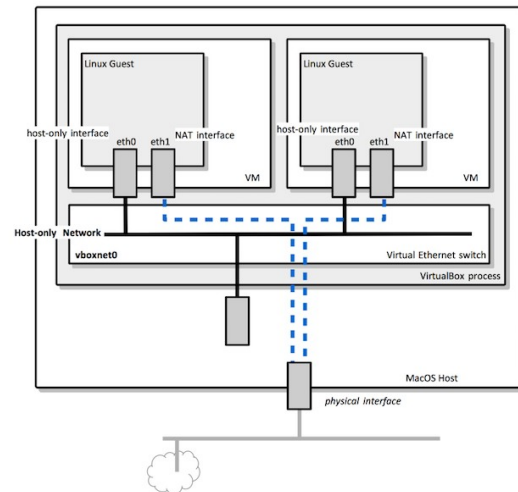
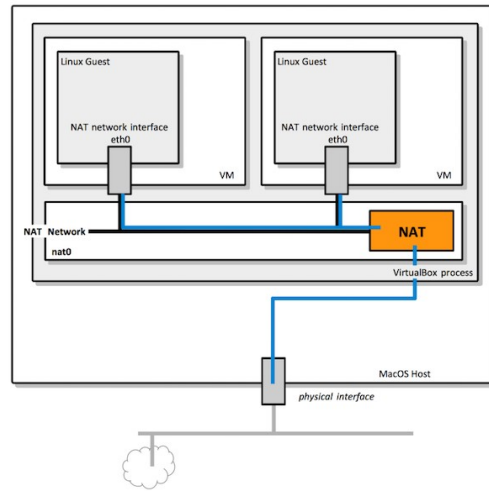
VirtualBox

DEMO

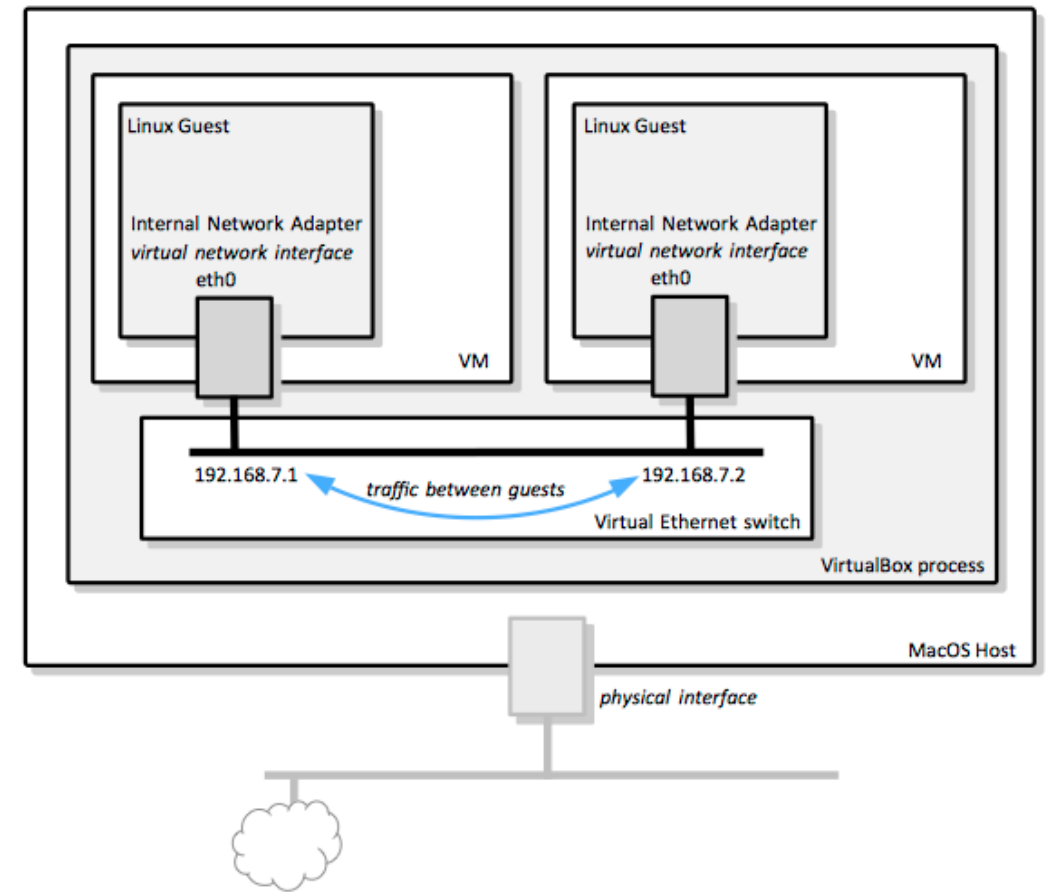


Virtualbox

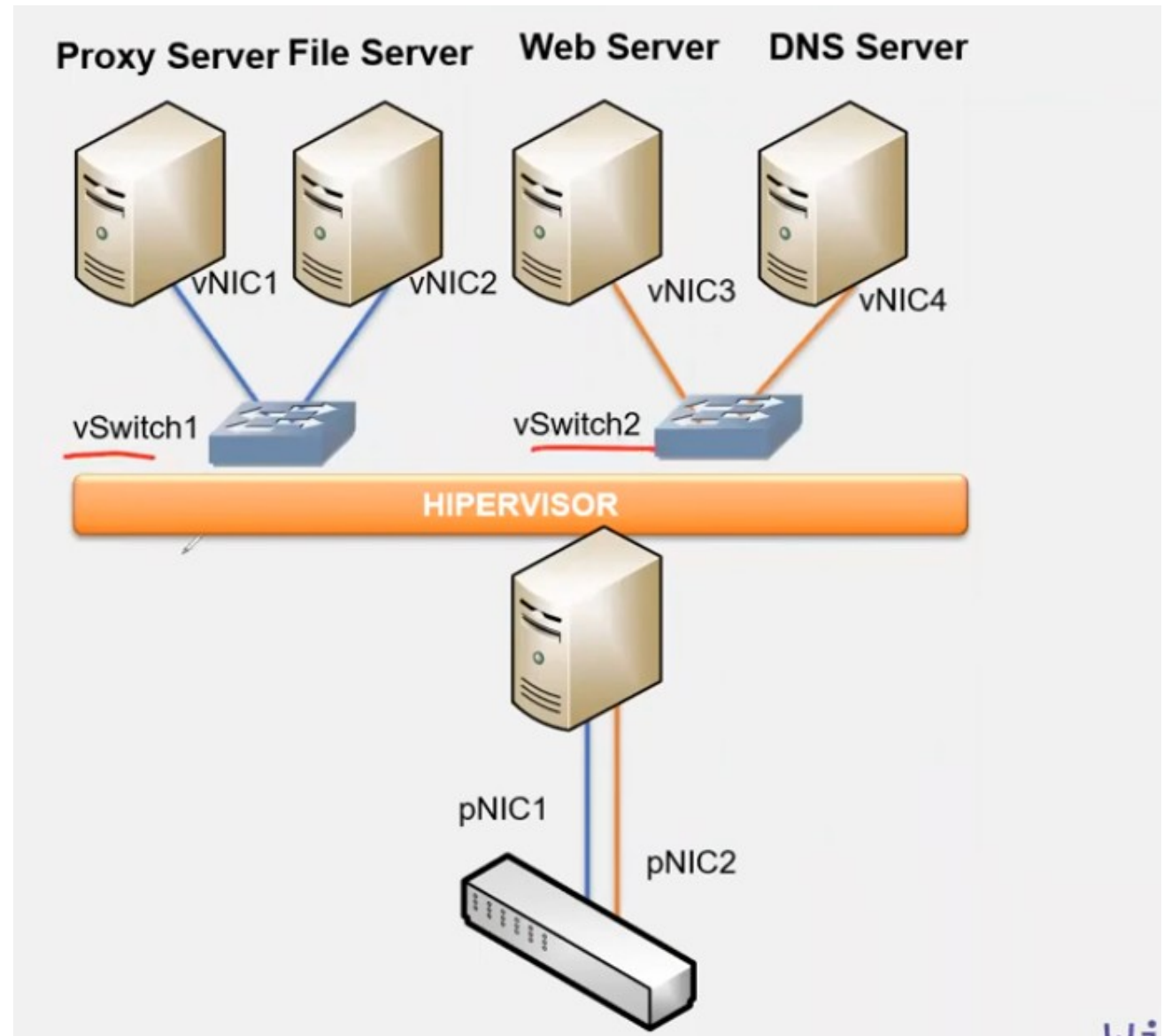
- Creación de maquinas virtuales
 - Plantillas
- Detener, suspender o migrar
- Instantáneas (snapshot)
- Importa o exportar
- Convertir MV
- CLI



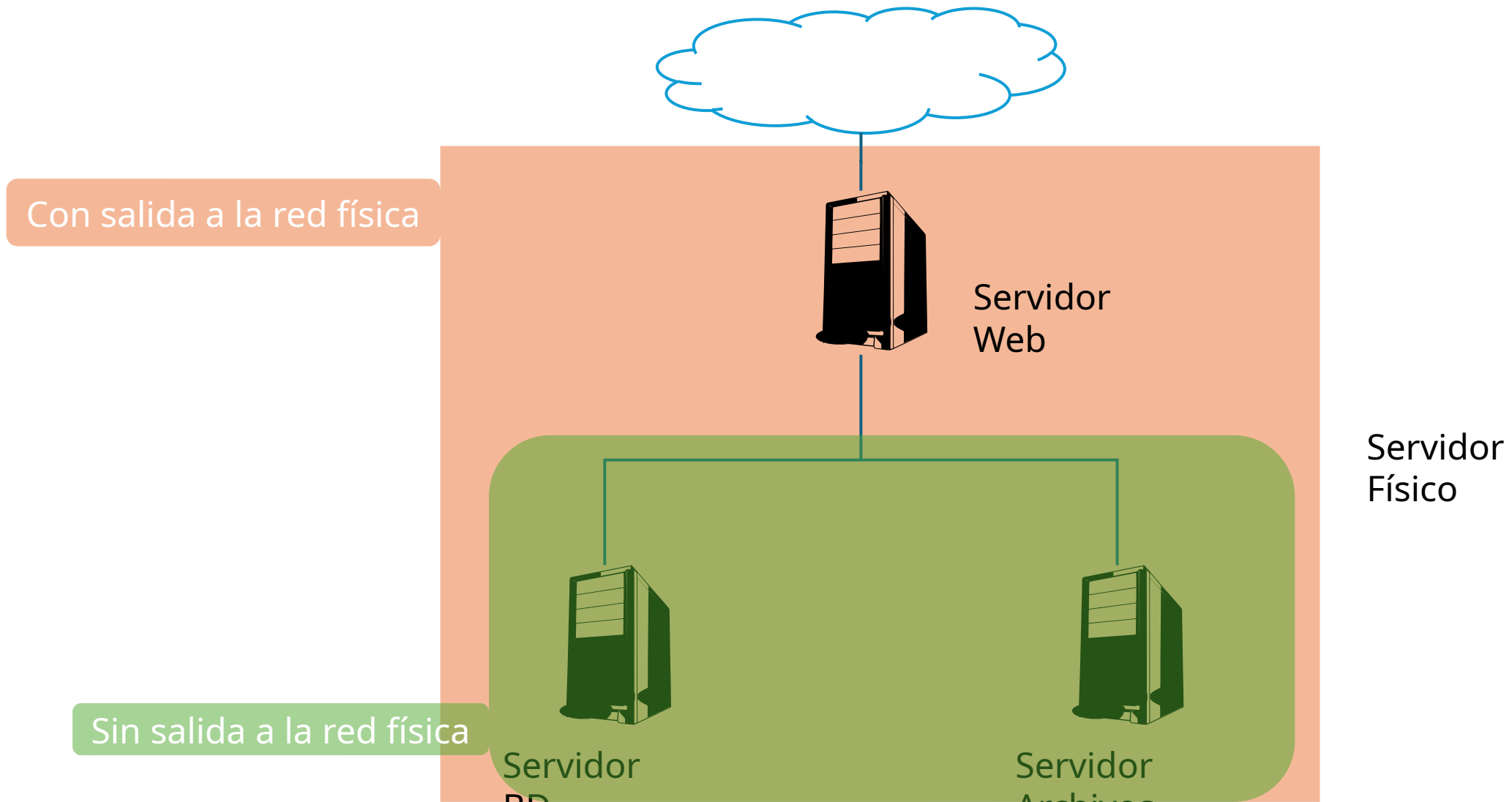
Internal Networking



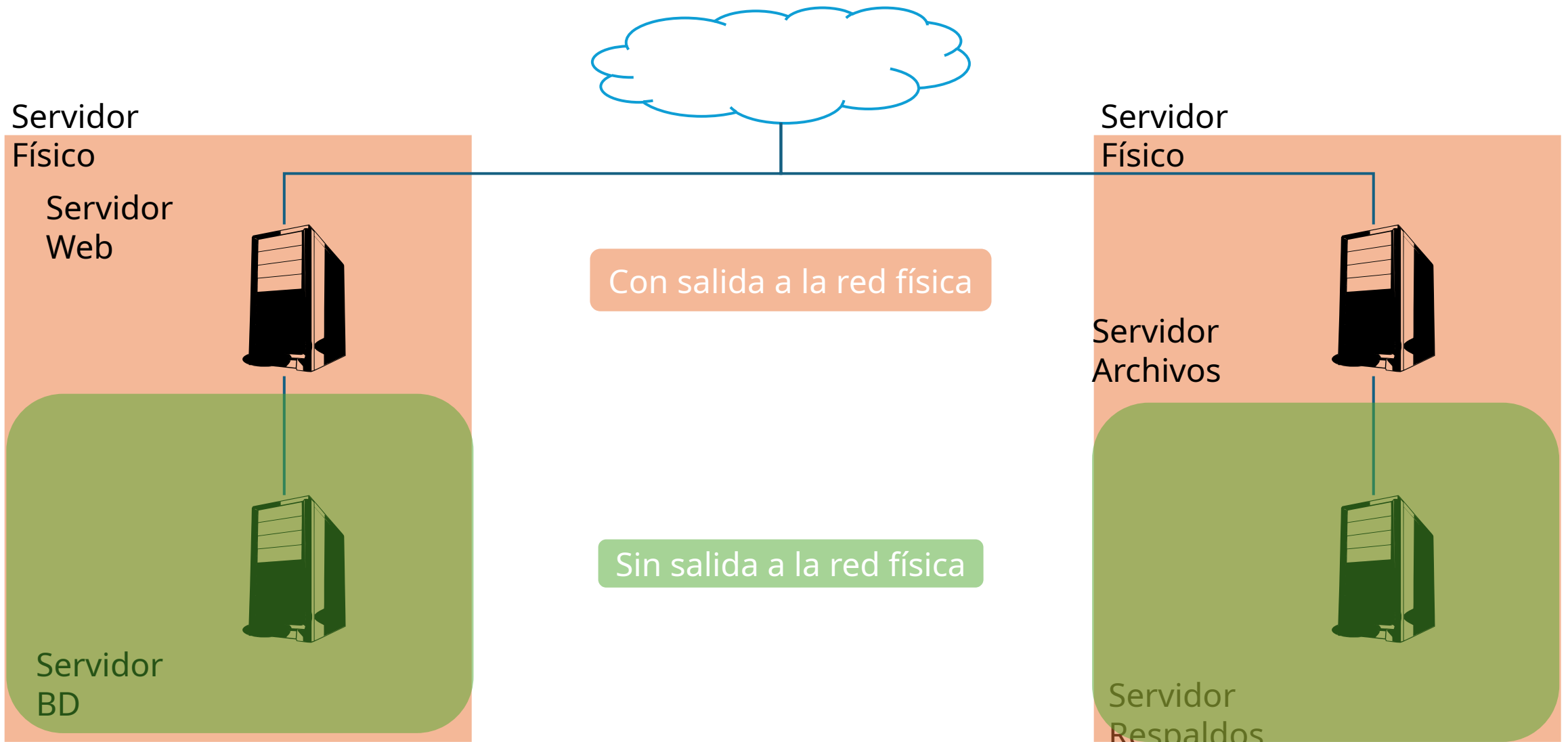
Virtualizació n de red a nivel MV



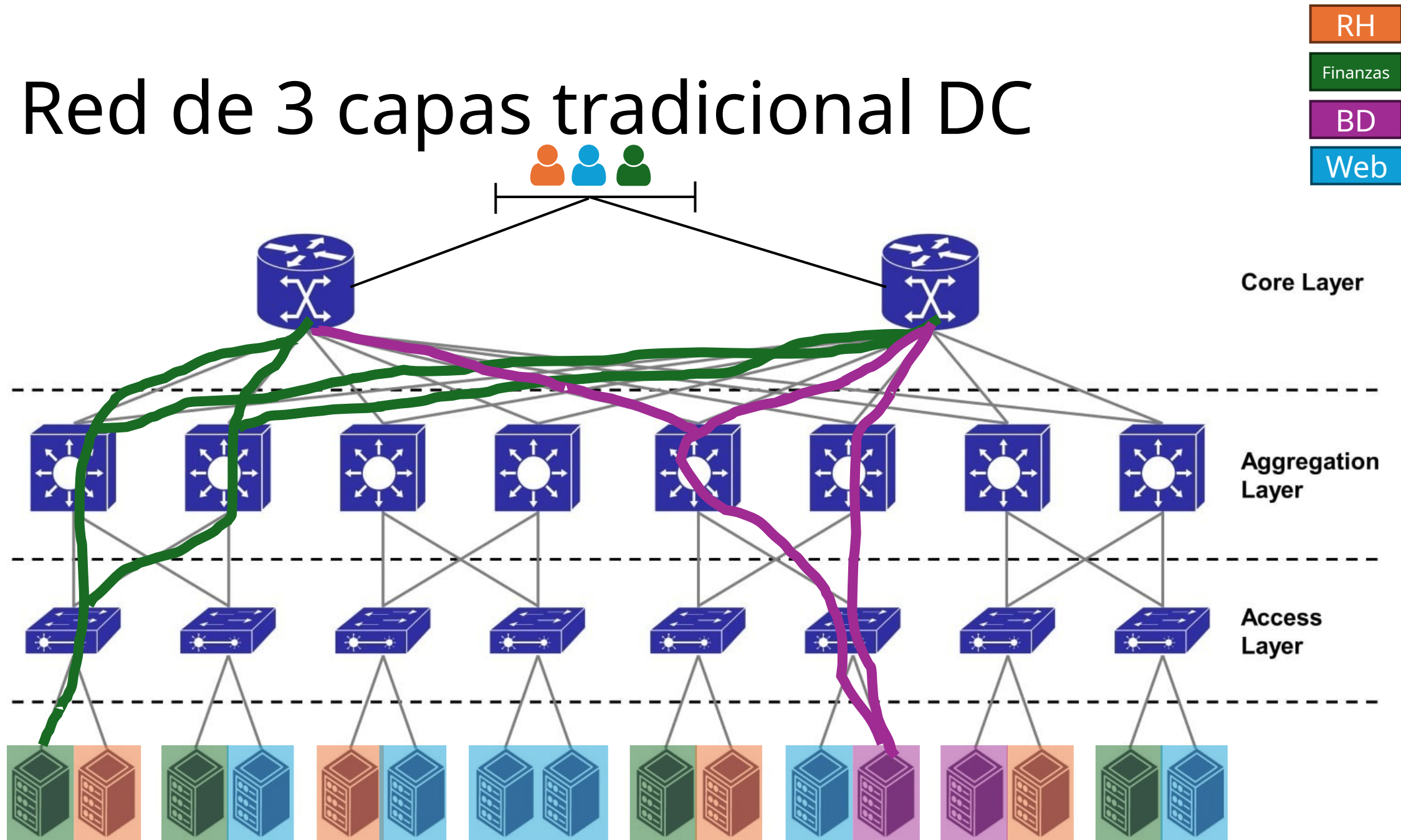
Ejercicio 1



Ejercicio 2

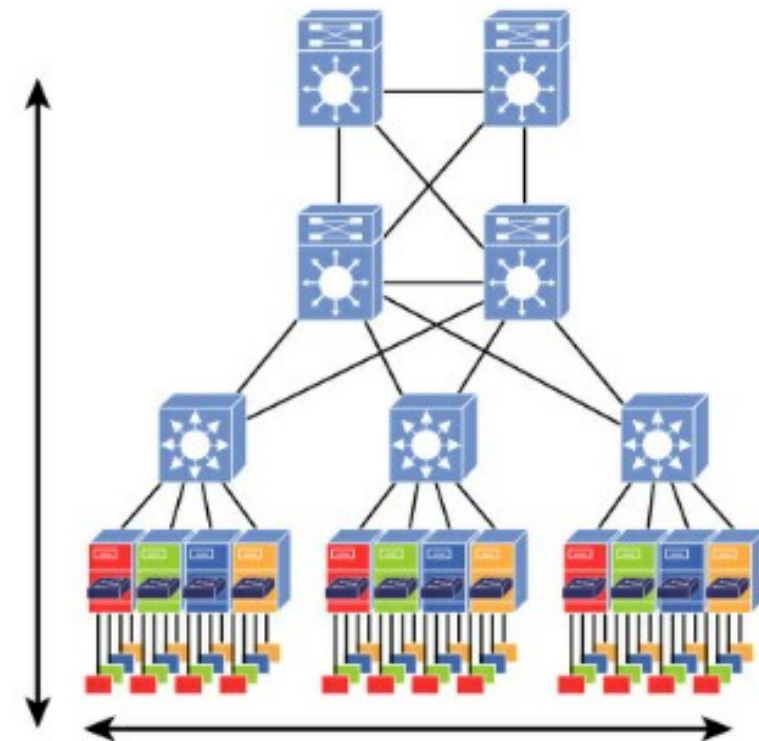


Red de 3 capas tradicional DC



Como evoluciona el tráfico

North-South traffic runs between the servers and the switches. Ex: Traffic from an external client to the app server.

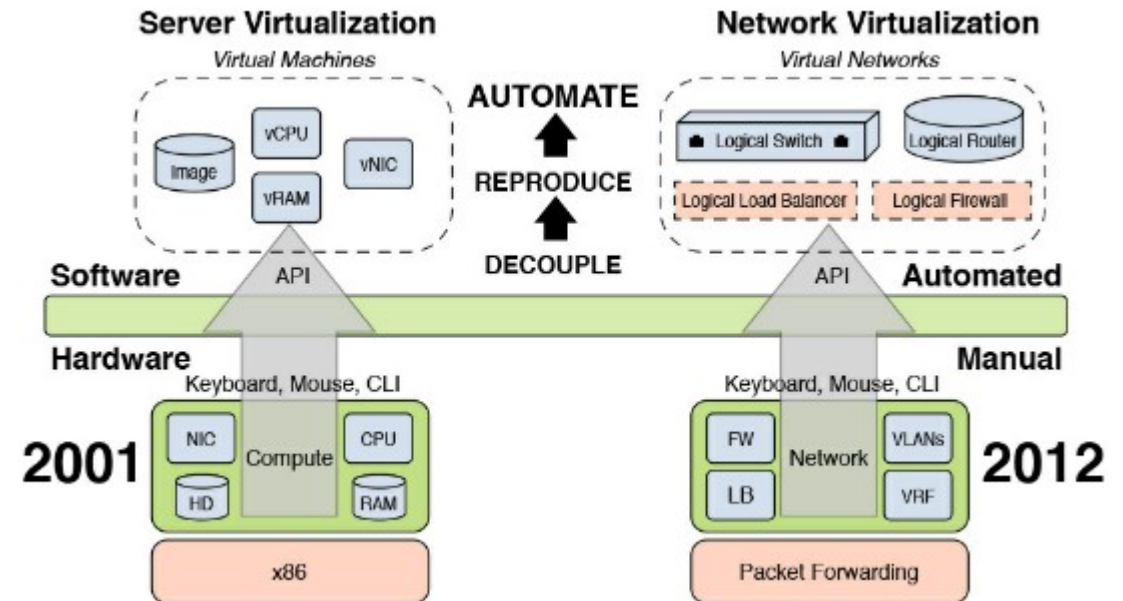


East-West traffic runs between Virtual Machines. In some cases this is on the same server but can be server to server as well.

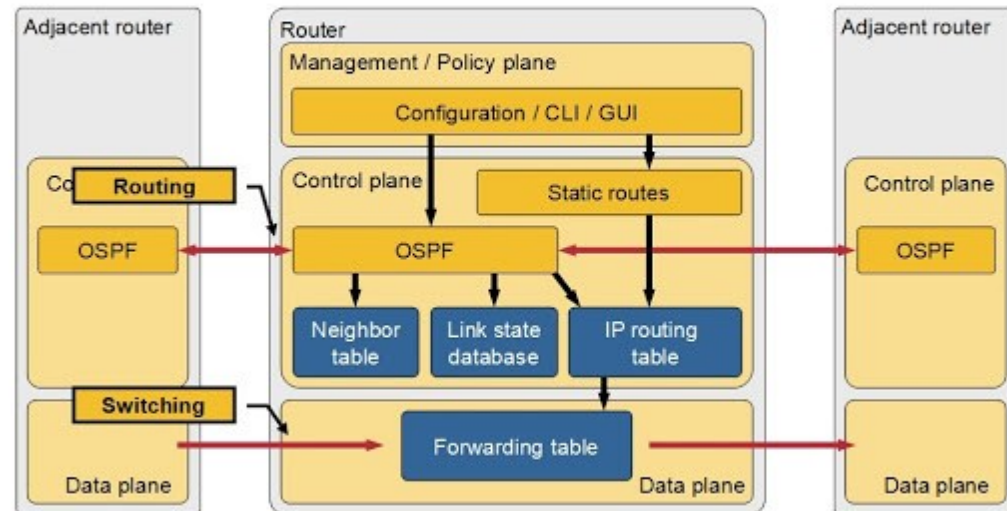
Figure 7-2 *In a virtual environment, VM-to-VM traffic (referred to as east-west traffic) makes up a significant portion of the overall traffic.*

Redes virtuales

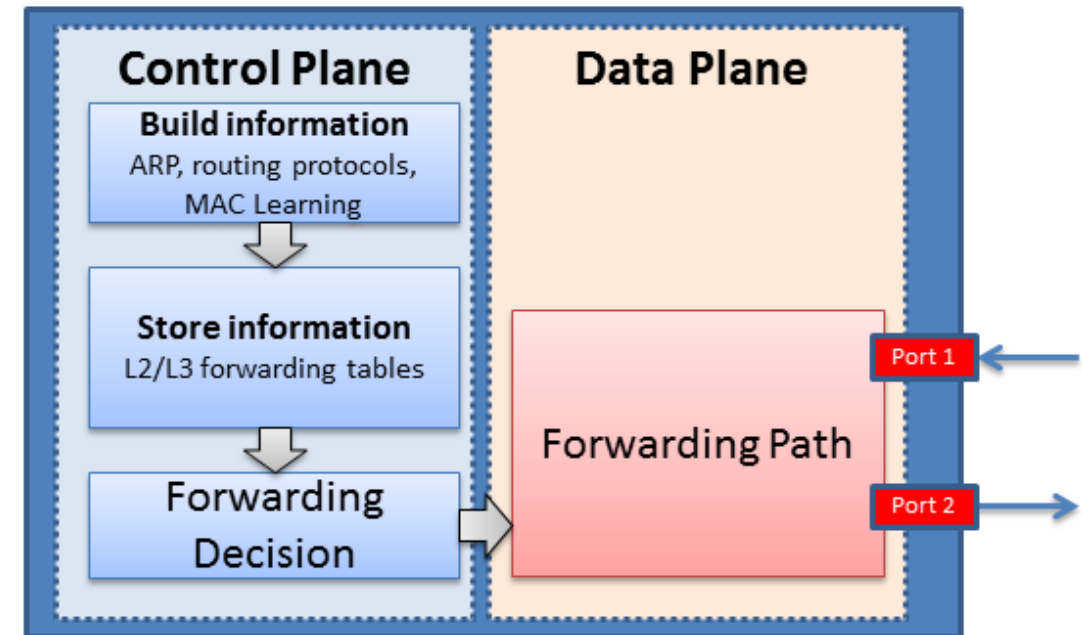
- Redes basadas en equipo de red
 - La virtualización se basa en las redes tradicionales y sus equipos
 - Representan una red lógica sobre una física
 - Interfaz virtual, VPN, LAG, VLAN, VRRP,
- Redes basadas en hipervisores
 - Utilizan la virtualización de cómputo para operar
 - Sus alcances dependen del hipervisor
- Software Defined Network
 - Hace una mezcla de las dos previas
 - Base para cloud



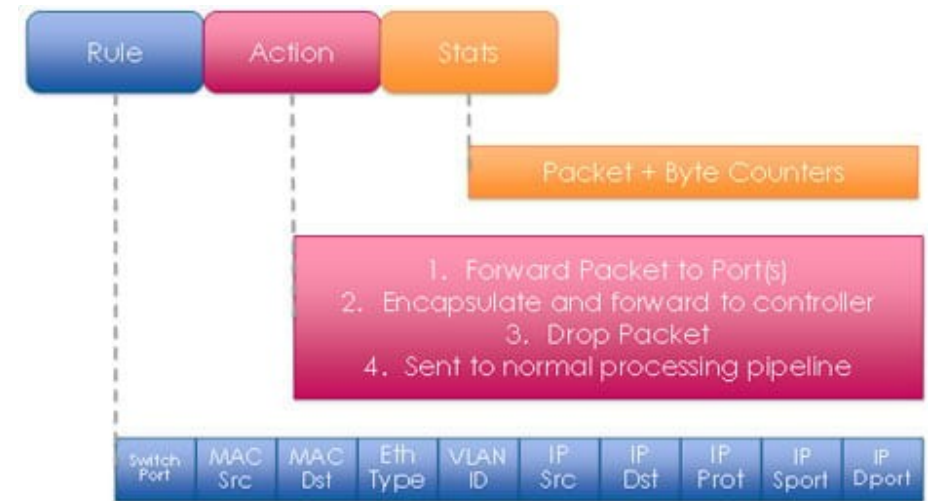
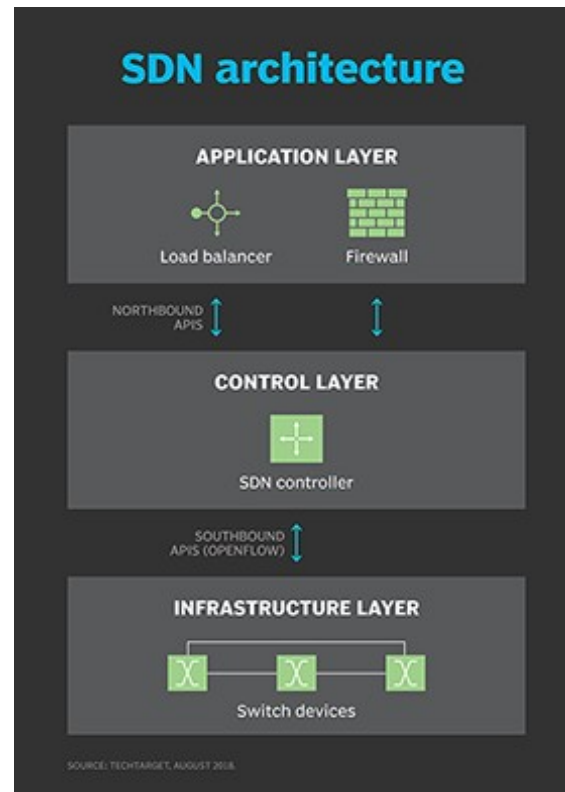
Management, Control and Data Planes



Switch



Openflow (Open Networking Foundation)

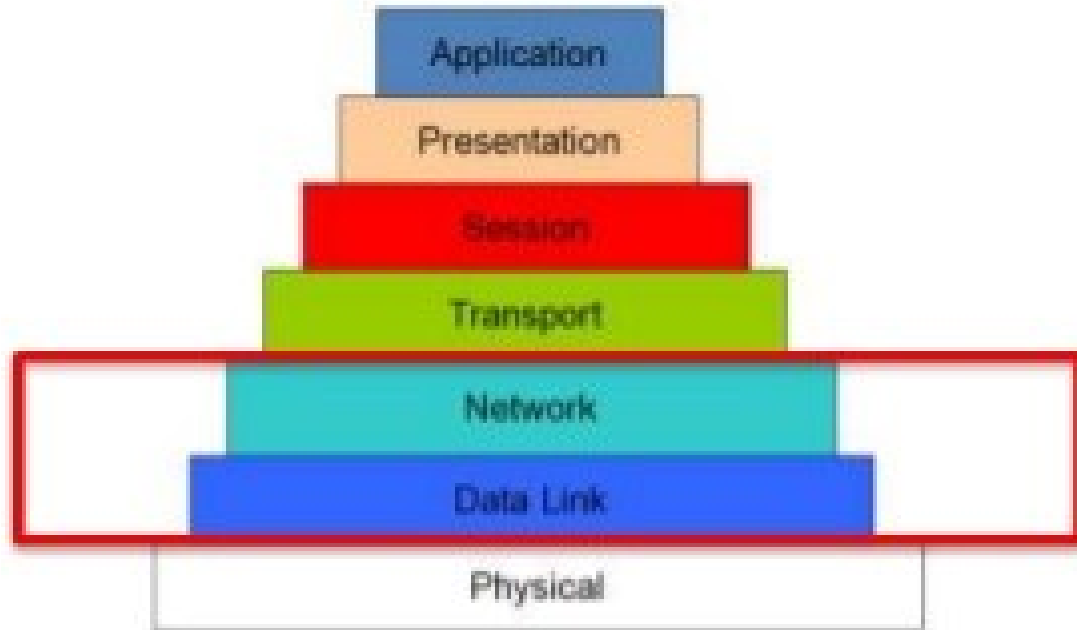


Virtualización d e Red. Introduc ción outube.com)



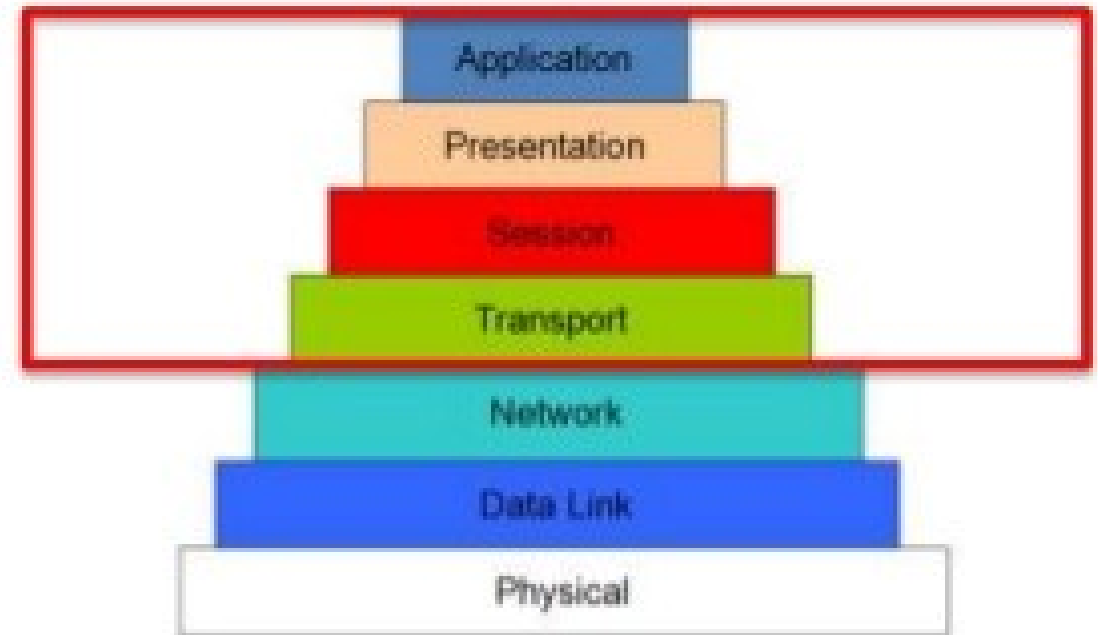
SDN vs NFV

SDN



- Optimize network infrastructure such as Ethernet switches, routers and wireless access points
- OSI Layer 2-3

NFV



- Optimize deployment of network functions such as: load balancer, firewall, WAN optimization controller, deep packet inspection etc.
- OSI Layer 4-7

Controladoras SDN

- OpenDayLight
 - NOX
 - Ryu
 - POX
 - ONOS
 - Floodlight
-
- CISCO ACI
 - Nuage Networks VSP
 - Juniper Contrail
 - Vmware NSX



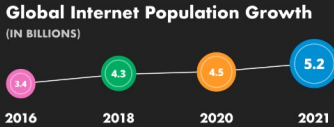
Data Never Sleeps 9.0

How much data is generated every minute?

The 2020 pandemic upended everything, from how we engage with each other to how we engage with brands and the digital world. At the same time, it transformed how we eat, how we work and how we entertain ourselves. Data never sleeps and it shows no signs of slowing down. In our 9th edition of the "Data Never Sleeps" infographic, we bring you a glimpse of how much data is created every digital minute in our increasingly data-driven world.



As of July 2021, the internet reaches 65% of the world's population and now represents 5.17 billion people—a 10% increase from January 2021. Of this total, 92.6 percent accessed the internet via mobile devices. According to Statista, the total amount of data consumed globally in 2021 was 79 zettabytes, an annual number projected to grow to over 180 zettabytes by 2025.



As the world changes, businesses need to change too—and that requires data. Domo gives you the power to make data-driven decisions at any moment, on any device, so that you can make smart choices in a rapidly changing world. Every click, swipe, share, or like tells you something about your customers and what they want, and Domo is here to help you and your business make sense of all of it.

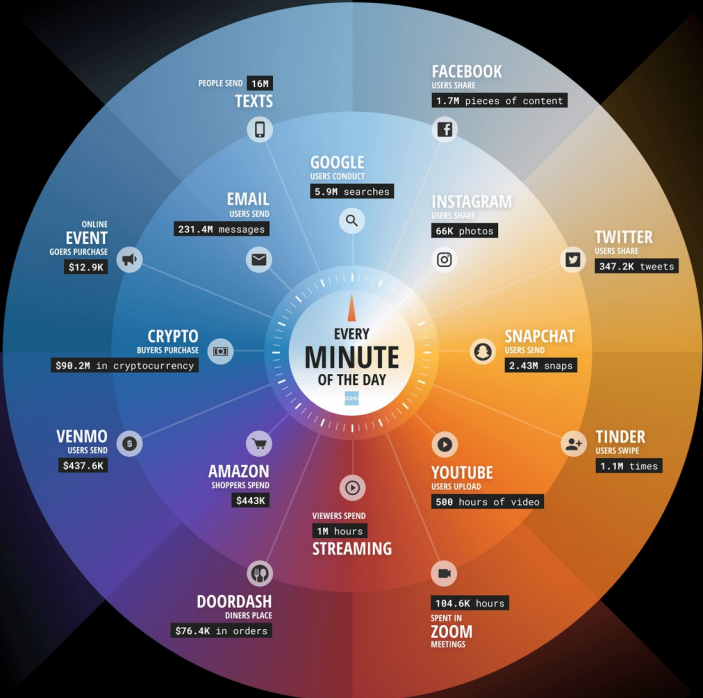
Learn more at domo.com

SOURCES: GLOBAL IQ, BUSINESS OF APPS, DUSTIN STOUT, HOOTSUITE, EXPANDED RAMBLINGS, INTERNET WORLD STATS, STATISTA, CNBC, BRANDWATCH, KILL THE CABLE BILL, YOUTUBE, KINISTA, THE VERGE, MANAGEMENT CONSULTING GROUP, A CASE ANALYSIS APPROACH, INTERNET LIVE STATS, SODA, STATISTA

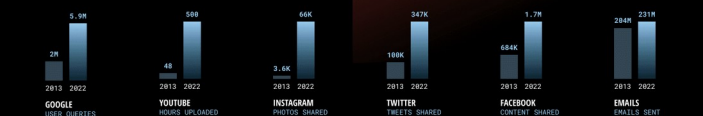


DATA NEVER SLEEPS 10.0

Over the last ten years, digital engagement through social media, streaming content, online purchasing, peer-to-peer payments and other activities has increased hundreds and even thousands of percentage points. While the world has faced a pandemic, economic ups and downs, and global unrest, there has been one constant in society: our increasing use of new digital tools to support our personal and business needs, from connecting and communicating to conducting transactions and business. In this 10th annual "Data Never Sleeps" infographic, we share a glimpse at just how much data the internet produces each minute from some of this activity, marveling at the volume and variety of information that has been generated.



DATA NEVER SLEEPS 1.0 VS. 10.0



To succeed in an increasingly digital world where the volume of data created keeps accelerating, businesses need the right tools to put that data to work right where work gets done. Domo gives you the power to rapidly unlock value from all your data, regardless of where it lives, and drive actions across your organization that will improve business outcomes. Every click, swipe, share, or like tells a story, and Domo helps you do something powerful with it.

LEARN MORE AT [DOMO.COM](https://domo.com)

SOURCES: Global Media Insights, Oberlo, Hootsuite, Earthen, Matthew Woodward.co.uk, Web Tribunal, Statista.com, Global IQ, Business of Apps, Query Genie, Young and Rubicam, Gearing Data, 888 World, Doordash, TechCrunch, Statista, Data Never Sleeps 1.0



Data Never Sleeps 11.0

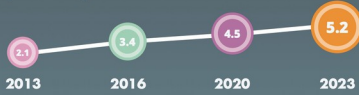
Domo has been keeping tabs on the world's data usage—in a minute—for over a decade now. What the numbers consistently show is that how we use data is always evolving—and that data isn't slowing down. We're also seeing some big changes. The rise of Artificial Intelligence (AI) is reshaping the way we communicate, work, and create. Digital payments continue to replace traditional transactions. Taylor Swift streams in countless headphones. And a rash of cybercrime grows alongside these digital experiences.

In Domo's 11th edition of Data Never Sleeps, we take the pulse of our digital age, where every click, swipe, and stream fuels an ever-expanding digital universe. These are not just numbers; they are the heartbeat of a world where data reigns supreme.



The world's internet population continues to grow significantly year-over-year. As of November 2023, the internet represents 5.2 billion people—approximately 64.6% of the global population. According to Statista, the total amount of data predicted to be created, captured, copied, and consumed globally in 2023 is 120 zettabytes, a number projected to grow to 181 zettabytes by 2025.

Global Internet Population Growth (in billions)



As data grows and evolves, businesses need to grow and evolve, too. Domo helps you harness the power of data so you can change as quickly as the world changes and make data-driven decisions that set you apart from the crowd. Let Domo help you make sense of all the clicks, swipes, and shares so you can see the big picture that a lot of small decisions make.

Learn more at domo.com

SOURCES: EARTHWEB, DUSTIN STOUT, DEMANDSAGE, HOOTSUITE, BUSINESSINSIDER, DOORDASH, SOCIAL.PLOT.X, TWITTER.COM, GITNEX, INVIGATE, THINKIMPACT, SIFMA.ORG, STATISTA, PR NEWSWIRE, NETSCOUT



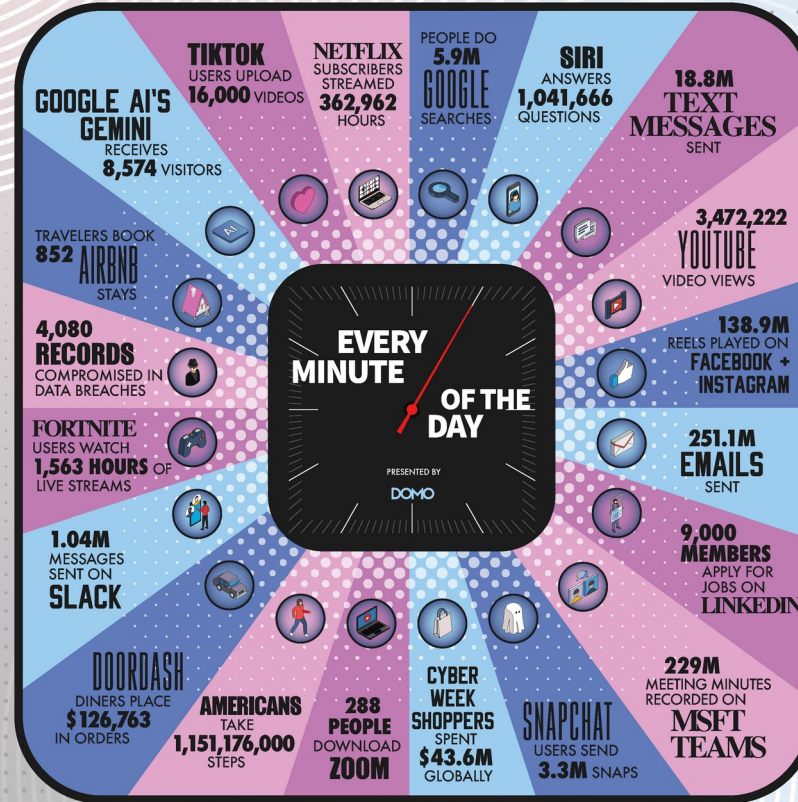
DATA NEVER SLEEPS



Every minute of every day, the world generates a dizzying amount of data, and how we interact with it is constantly changing. AI tools are now answering millions of questions in real time and are transforming how we work, shop, and connect. Digital platforms are seeing explosive usage, with billions of emails, texts, and reels shared every day. Entertainment continues to drive engagement across streaming, gaming, and social media while e-commerce is setting new benchmarks as digital habits evolve and expand at an unprecedented pace.

In Domo's 12th edition of Data Never Sleeps, we capture a snapshot of this world powered by the rapid rise of data, AI, and digital activity, shaping every moment of modern life.

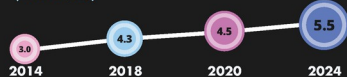
12



The world's internet population continues to grow significantly year-over-year. As of late 2024, 5.52 billion people—approximately 67.5% of the global population—are online.

According to industry analysts, the total amount of data created, captured, copied, and consumed globally is expected to reach 149 zettabytes by the end of 2024, with projections surpassing 394 zettabytes by 2028.

Global Internet Population Growth (IN BILLIONS)



As the volume and complexity of data accelerates, business success increasingly depends on the ability to turn information into insights. Domo helps you harness the power of data and AI so you can adapt as quickly as the world changes and make data-driven decisions that set you apart. Let Domo help you make sense of all the clicks, swipes, and streams so you can see the big picture shaped by every small decision.

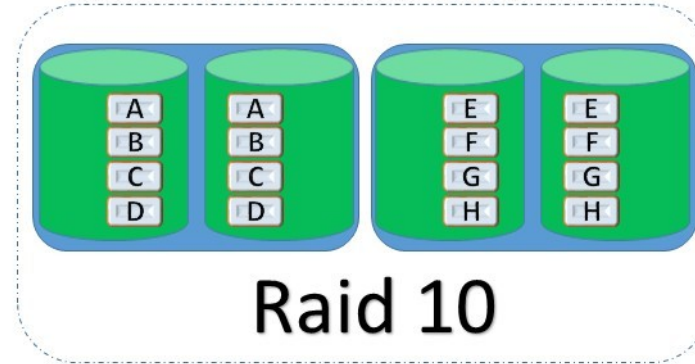
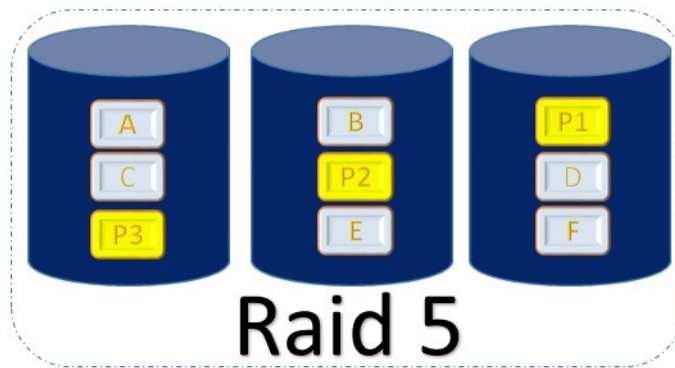
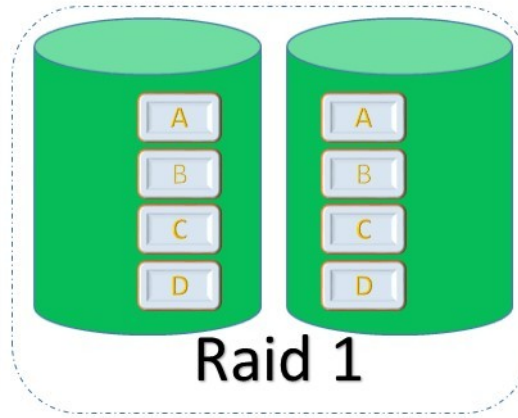
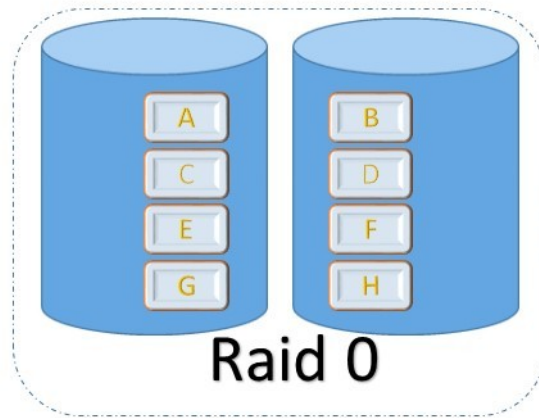
Learn more at domo.com

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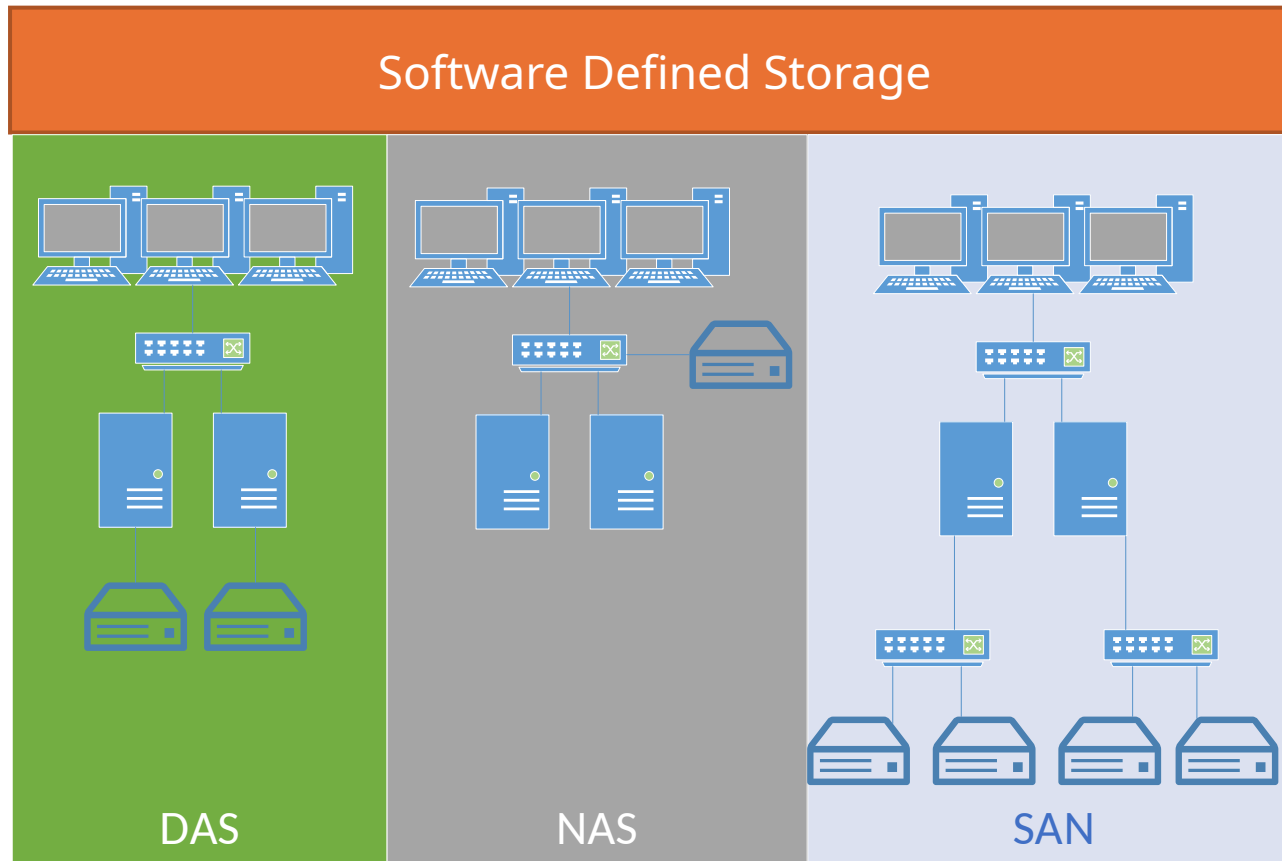


Virtualización de almacenamiento

Arreglos de discos



Almacenamiento Virtual



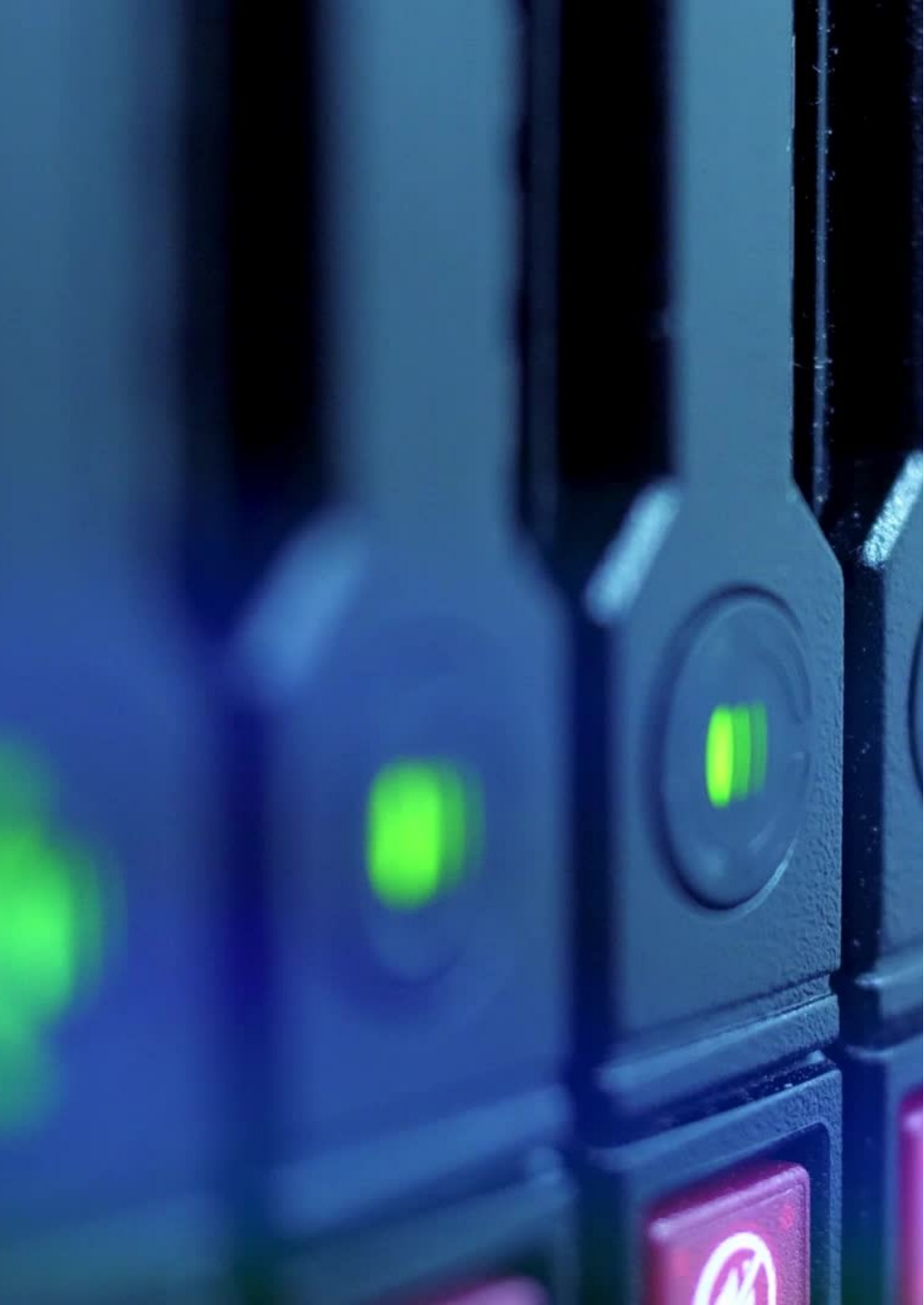
- SV prove grupos de almacenamiento estructurados en niveles y presentados en el medio y protocolo adecuado.
- SDS además debe:
 - **Automatización**
 - **Interface estándar API**
 - **Ruta de datos virtualizada** (Block, File and/or Object interfaces)
 - **Escalabilidad**
 - **Transparencia**



Storage Virtualization and Software Defined Storage (youtube.com)



Jeffrey Lush
Storage Virtualization vs Software Defined Storage

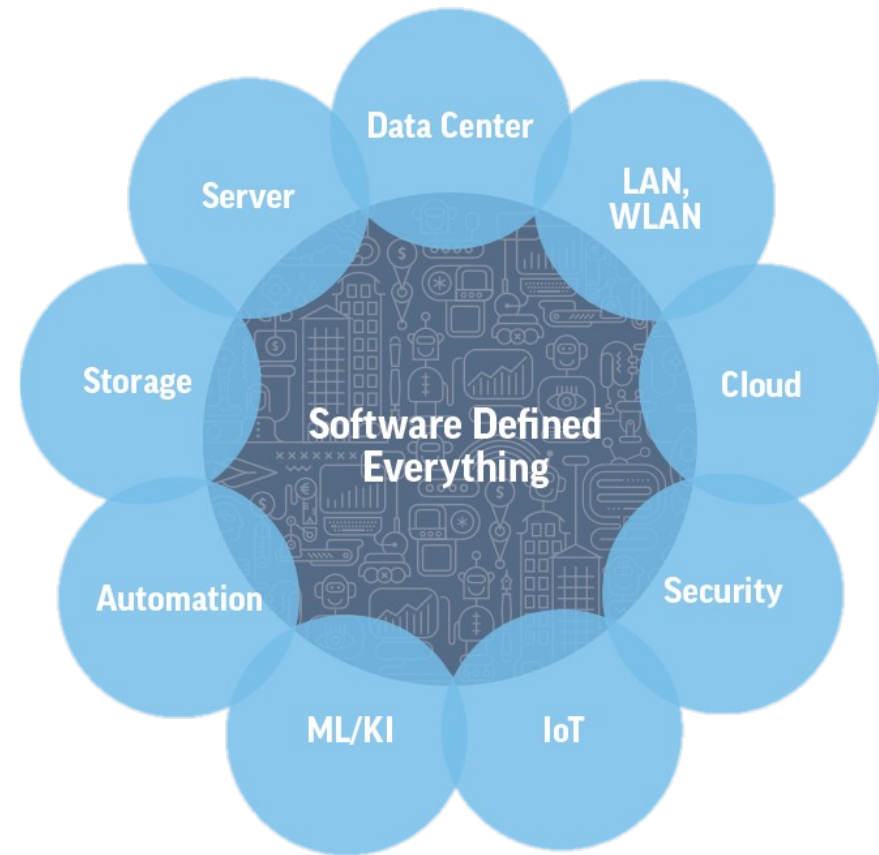
A close-up, slightly blurred photograph of a server rack. Several vertical server units are visible, each with a glowing green light indicator. The lighting is dim, with the green lights providing the primary illumination. The units are dark-colored, possibly black or dark blue.

Tipos de almacenamiento

- File-base storage
 - NAS
 - SMB, NFS, CIFS
- Block-base storage
 - SAN
 - SCSI, iSCSI, FC
- Object-base storage
 - Dualidad, datos y metadatos

SDx (Software Defined Everything)

- SDC (Software Defined Computer)
- SDN (Software Defined Network)
- SDS (Software Defined Storage)
- SDWAN (Software Defined WAN)
- SDSec (Software Defined Security)
- SDDC (Software Defined Data Center)



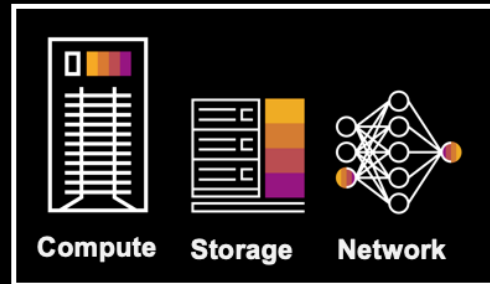
What is a Hyper-Converged Infrastructure?

Traditional



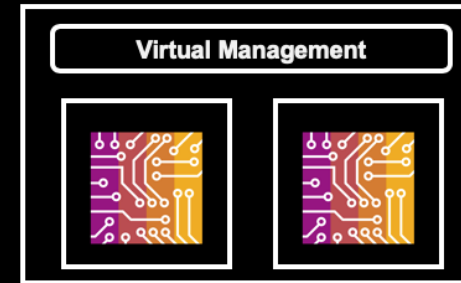
- Physically distinct components
- Different vendors
- Customers manage compatability
- 3 teams to operate

Converged



- Distinct components
- Pre-built and validated by one vendor
- Simplified support
- Less in-depth technical knowledge per layer

Hyper-Converged



- Software-defined (Virtualized HW components)
- Centralized management of pooled resources
- Easy to deploy and operate
- Highly scalable and sharable resources
- One vendor: end-to-end

Beneficios para la seguridad

- Aislamiento de MV
- Snapshots para regresar después de una infección durante una evaluación
- Micro segmentación de la red, evita movimientos laterales
- Mejora la recuperación de desastres
- Aplicación en Sandboxing, Honeypots y VDI